Embryology of head & neck

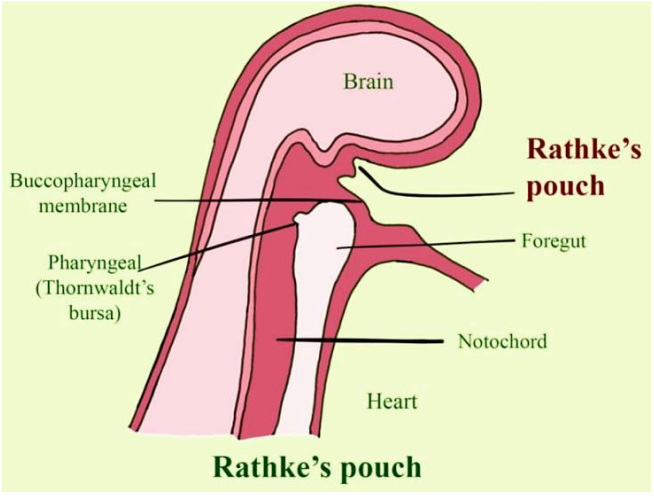
Pharyngeal arches, pouches and clefts; development of the ear, face, palate, tongue and thyroid, and the congenital anomalies each one explains.

1. Pharyngeal apparatus
2. Development of the ear
3. Development of the face
4. Development of the palate
5. Development of the tongue
6. Development of the thyroid and parathyroids
7. Development of the nose and sinuses

01 Pharyngeal apparatus

- From the 4th week: **arches** (mesoderm and neural crest, each with a nerve, artery, cartilage and muscle), **pouches** (endoderm, inside) and **clefts** (ectoderm, outside), separated by membranes.
- Five arches develop: **1st, 2nd, 3rd, 4th and 6th** (the 5th regresses).
- **Nerves:** 1st V3, 2nd VII, 3rd IX, 4th superior laryngeal (X), 6th recurrent laryngeal (X).
- **Clefts:** the 1st becomes the **EAC**; the 2nd–4th are buried by the overgrowing 2nd arch as the **cervical sinus**, which normally disappears.
- **Rathke's pouch** (from the stomodeum roof) forms the anterior pituitary.

> **CLINICAL PEARL** A persistent cervical sinus or tract gives a branchial cyst, sinus or fistula; its relation to nerves and arteries tells you which arch it came from.



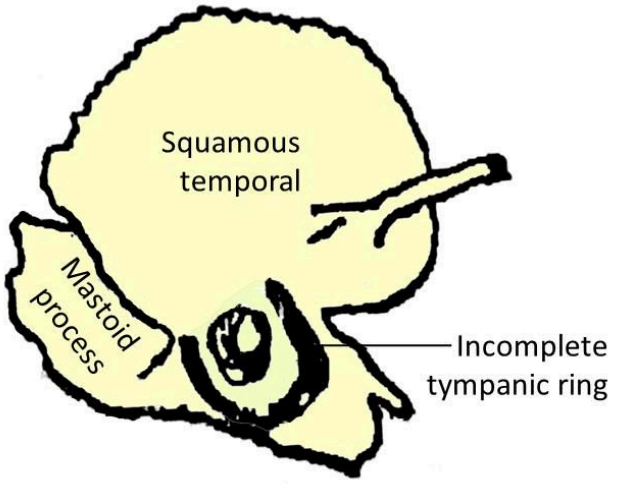
Rathke's pouch and the buccopharyngeal membrane
ENT Steps Anatomy, p. 202

02 Development of the ear

- **Inner ear:** from the **otic placode** (ectoderm) → otic pit → **otic vesicle** (week 4). The membranous labyrinth is adult size by about week 20–25; the otic capsule ossifies from 14 centres.
- **Middle ear and ET:** from the **1st pharyngeal pouch** (tubotympanic recess).
- **Ossicles: 1st arch** gives the malleus head and incus body (and short process); **2nd arch** gives the handle of malleus, long process of incus and stapes superstructure; the footplate is partly from the otic capsule.

- **EAC:** from the **1st cleft**; a meatal plug canalises from about week 21–28.
- **Auricle:** from six **hillocks of His** (three on the 1st arch, three on the 2nd).
- **TM:** ectoderm (1st cleft), mesoderm and endoderm (1st pouch).
- **At birth:** tympanic ring incomplete, no mastoid process (facial nerve exits superficially), antrum present and adult-sized.

CLINICAL PEARL Inner ear develops separately from the outer and middle ear, so microtia with canal atresia often has a normal cochlea. In infants the facial nerve is superficial at the stylomastoid foramen, so a postaural incision must stay high.

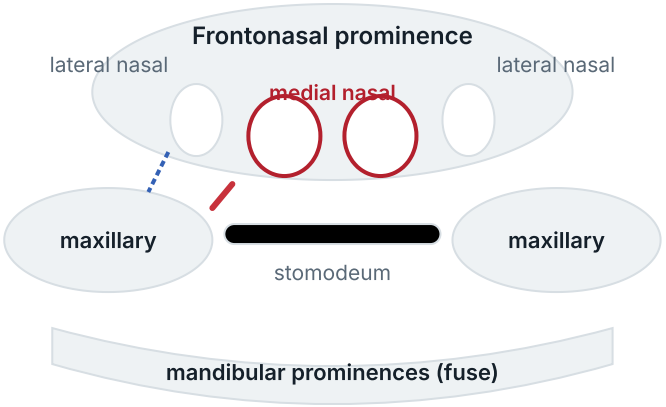


Temporal bone at birth: incomplete tympanic ring, no mastoid process
Dr. Sameh Doss handout, p. 37

03 Development of the face

- Five prominences around the stomodeum by week 4–5: one **frontonasal** and paired **maxillary** and **mandibular** prominences.
- Nasal placodes sink to form **medial and lateral nasal prominences**.
- **Medial nasal prominences** fuse to form the **intermaxillary segment**: philtrum, premaxilla with four incisors, primary palate.
- The **maxillary prominence** fuses with the medial nasal prominence to form the upper lip; the **nasolacrimal groove** between the maxillary and lateral nasal prominences forms the nasolacrimal duct.
- **Mandibular prominences** fuse to form the lower lip and jaw.

CLINICAL PEARL Cleft lip results from failure of the maxillary and medial nasal prominences to fuse, so it lies lateral to the philtrum.



Red: cleft lip site (maxillary + medial nasal fail to fuse)
Blue dashed: nasolacrimal groove (maxillary + lateral nasal)
Medial nasal prominences fuse: philtrum, premaxilla, primary palate

Facial prominences at about week 5. Cleft lip forms where the maxillary and medial nasal prominences fail to fuse, beside the philtrum.

04 Development of the palate

- **Primary palate** (premaxilla) from the intermaxillary segment, in front of the **incisive foramen**.
- **Secondary palate** from the two **palatal shelves** of the maxillary prominences, which grow down beside the tongue, rise above it and fuse in the midline (weeks 7–10), front to back.
- The nasal septum grows down and fuses with the palate.
- **Cleft lip and palate** classification: anterior to the incisive foramen (lip, alveolus), posterior to it (secondary palate), or both. **Submucous cleft:** bifid uvula, notch in the hard palate, translucent midline (zona pellucida).

> **CLINICAL PEARL** Submucous cleft is a contraindication to adenoidectomy: removing the adenoid pad can cause velopharyngeal insufficiency and hypernasal speech.

05 Development of the tongue

- **Anterior two-thirds:** from the 1st arch (two **lateral lingual swellings** and the median **tuberculum impar**) → general sensation by **V3** (lingual).
- **Posterior third:** from the **hypobranchial eminence** (mainly 3rd arch), which overgrows the 2nd-arch copula → **IX**.
- **Most posterior part and epiglottis:** 4th arch → **X**.
- **Muscles:** from occipital myotomes → **XII** (except palatoglossus, X).
- The **sulcus terminalis** marks the junction; the **foramen caecum** at its apex is where the thyroid began.

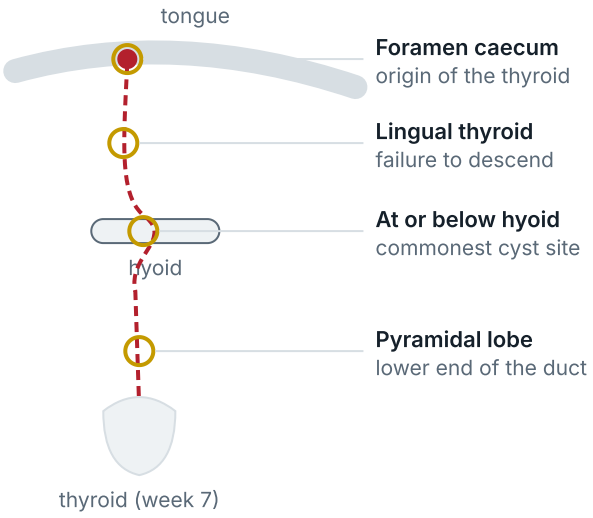
> **CLINICAL PEARL** Taste in the anterior two-thirds is VII (chorda tympani) because the 2nd-arch nerve invades early, even though the mucosa is 1st-arch.

06 Development of the thyroid and parathyroids

- Thyroid starts at the **foramen caecum** (week 3–4) as an endodermal diverticulum, descends in front of the pharynx and the **hyoid**, attached by the **thyroglossal duct**, reaching the front of the trachea by week 7.

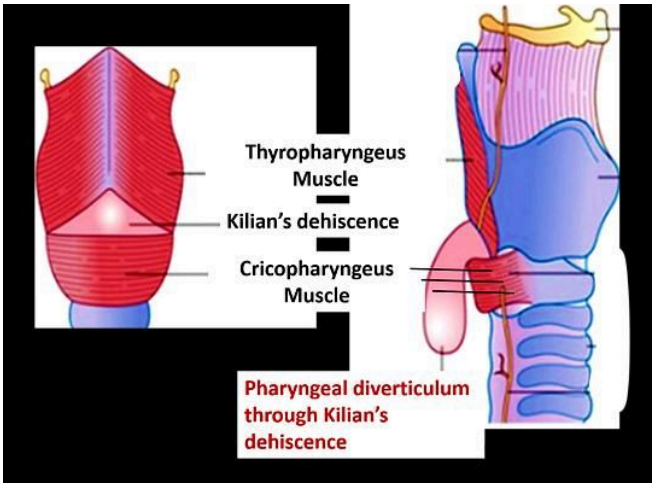
- The duct normally disappears by week 10; its lower end may persist as the **pyramidal lobe**.
- **C cells** come from the **ultimobranchial body** (4th/5th pouch).
- **Inferior parathyroids and thymus:** 3rd pouch. **Superior parathyroids:** 4th pouch.
- **Anomalies:** thyroglossal duct cyst (midline, moves with swallowing and tongue protrusion), **lingual thyroid** (failure to descend; may be the only thyroid tissue), ectopic thyroid anywhere along the tract.

> **CLINICAL PEARL** Before excising a lingual thyroid or thyroglossal cyst, confirm normal thyroid tissue in the neck with ultrasound or a scan.



Sistrunk's operation removes the middle of the hyoid with the tract.

Thyroglossal tract. Remnants along the path explain the midline neck cyst that rises with tongue protrusion, and why the hyoid body is removed with it.



Killian's dehiscence and pharyngeal pouch
ENT Steps Anatomy, p. 198

07 Development of the nose and sinuses

- Nasal placodes → nasal pits → the oronasal membrane ruptures to form the **primitive choanae** (week 6–7). Failure gives **choanal atresia**.
- Lateral nasal wall develops ridges (ethmoturbinals) that become the turbinates; the furrows between them form the meatuses and sinus outgrowths.
- Sinuses arise as outpouchings of the nasal mucosa: maxillary (3rd month in utero), ethmoid (4th–5th month), sphenoid and frontal after birth.

- **Choanal atresia:** about 90% bony or mixed and 10% membranous; unilateral is more common. Bilateral atresia causes cyclical cyanosis relieved by crying in newborns (obligate nasal breathers). Associated with **CHARGE**.

> CLINICAL PEARL Failure to pass a size 6 catheter through the nose in a newborn with breathing difficulty suggests choanal atresia; confirm with CT.

Tables

Pharyngeal arch derivatives

Arch	Nerve	Skeletal	Muscles
1st	V3	Malleus (head), incus (body), Meckel's cartilage, sphenomandibular ligament	Mastication, tensor tympani, tensor veli palatini, mylohyoid, anterior digastric
2nd	VII	Stapes (superstructure), styloid, stylohyoid ligament, lesser horn and upper hyoid body	Facial expression, stapedius, stylohyoid, posterior digastric
3rd	IX	Greater horn and lower hyoid body	Stylopharyngeus
4th	X (superior laryngeal)	Thyroid cartilage	Cricothyroid, pharyngeal constrictors, levator veli palatini
6th	X (recurrent laryngeal)	Cricoid, arytenoid, corniculate	Intrinsic laryngeal muscles (except cricothyroid)

Pouches: 1st → middle ear and ET; 2nd → tonsillar fossa; 3rd → inferior parathyroid and thymus; 4th → superior parathyroid and C cells.

Branchial cleft anomalies

Type	Opening / site	Tract relations
1st (Work type I)	Near the EAC; ectoderm only	Duplication of the EAC, lateral to VII
1st (Work type II)	Angle of mandible; ectoderm + mesoderm	Through the parotid, close to the facial nerve
2nd (commonest)	Anterior border of SCM, lower third	Between ICA and ECA, over IX and XII, to the tonsillar fossa
3rd	Anterior border of SCM, lower	Behind the ICA, pierces thyrohyoid membrane above the internal laryngeal nerve, to the piriform sinus
4th	Usually left; piriform sinus apex	Below the superior laryngeal nerve; recurrent left neck abscess or thyroiditis

Pharyngeal pouch derivatives

Pouch	Derivative
1st	Middle ear cavity, Eustachian tube, mastoid antrum
2nd	Tonsillar fossa (crypta magna), palatine tonsil epithelium
3rd	Inferior parathyroid (dorsal), thymus (ventral)
4th	Superior parathyroid
4th/5th (ultimobranchial body)	Parafollicular C cells

Development of the ear

Part	Origin	Timing / notes
Inner ear	Otic placode → otic vesicle	Week 4; adult size by ~week 20–25
Middle ear and ET	1st pouch (tubotympanic recess)	Pneumatisation continues after birth
Malleus head, incus body	1st arch	—
Handle of malleus, long process of incus, stapes superstructure	2nd arch	Footplate partly otic capsule
EAC	1st cleft (meatal plug canalises)	Weeks 21–28
Auricle	Six hillocks of His (1st and 2nd arches)	Weeks 6–8
Mastoid process	Grows after birth	Absent at birth; facial nerve superficial

Cleft lip and palate: anatomy

Type	Structures involved	Cause
Cleft lip	Lip ± alveolus, in front of incisive foramen	Maxillary + medial nasal prominence failure
Cleft palate (secondary)	Behind incisive foramen	Palatal shelves fail to fuse
Complete cleft lip and palate	Both, through the alveolus	Both of the above
Submucous cleft	Bifid uvula, hard palate notch, zona pellucida	Muscle not united under intact mucosa

M Mnemonics

Arch nerves

5, 7, 9, 10, 10

1st V3, 2nd VII, 3rd IX, 4th SLN, 6th RLN

Pouch derivatives

Ear, Tonsil, Bottom-to-top parathyroids, C cells

1st middle ear, 2nd tonsil, 3rd inferior parathyroid + thymus, 4th superior parathyroid, ultimobranchial C cells

CHARGE

Coloboma, Heart, Atresia choanae, Retardation, Genital, Ear

Association with choanal atresia

Q Rapid recall

1. **Nerve of each pharyngeal arch?** 1st V3, 2nd VII, 3rd IX, 4th superior laryngeal (X), 6th recurrent laryngeal (X).
2. **Derivatives of the 1st pharyngeal pouch?** Middle ear cavity and Eustachian tube.
3. **Derivatives of the 2nd pouch?** Tonsillar fossa (crypta magna).
4. **Derivative of the 1st cleft?** External auditory canal.
5. **Origin of the inner ear?** Otic placode → otic pit → otic vesicle.
6. **Origin of the auricle?** Six hillocks of His, three from the 1st arch and three from the 2nd.
7. **Which prominences fail to fuse in cleft lip?** Maxillary and medial nasal prominences.
8. **Landmark separating primary and secondary palate?** Incisive foramen.

9. **Signs of a submucous cleft palate?** Bifid uvula, notch in the posterior hard palate, translucent midline (zona pellucida).
10. **Embryological origin of the posterior third of the tongue?** 3rd arch (hypobranchial eminence), so supplied by IX.
11. **Where does the thyroid originate?** Foramen caecum of the tongue.
12. **Proportion of bony vs membranous choanal atresia?** About 90% bony or mixed, 10% membranous.
13. **Syndrome associated with choanal atresia?** CHARGE (Coloboma, Heart, Atresia choanae, Retardation, Genital, Ear anomalies).
14. **Which tract does a 2nd branchial fistula follow?** Between the ICA and ECA, over IX and XII, to the tonsillar fossa.
15. **Ear structures at birth that are adult-sized?** Ossicles, middle ear cavity, antrum and inner ear.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.